LLM-Based Analysis of SEC Form 8-K Filings for New Product Identification (Assignment 2 - CAP5619)

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Repository: https://github.com/maufcost/sec-8k-llm-product-analysis

This project investigates the application of Large Language Models (LLMs) and Named Entity Recognition (NER) to automatically identify new product announcements within SEC Form 8-K filings. Publicly traded companies submit Form 8-K reports to disclose material corporate events, including financial results, acquisitions, management changes, and other developments.

The objective here was to develop an automated Python pipeline capable of retrieving SEC filings, processing unstructured financial documents, identifying newly announced products, and exporting the extracted information into a CSV dataset.

SEC filing data was retrieved through publicly available SEC EDGAR endpoints using Python (**requests**).. The company ticker reference file (**company_tickers.json**) was used to identify each company's Central Index Key (CIK). The SEC Submissions API was subsequently queried to obtain recent Form 8-K filings, including filing timestamps, accession numbers, and document references.

The initial analysis focused on NVIDIA (NVDA), followed by an expanded collection including AMD and Microsoft (MSFT). The last confirmed dataset contained 20 documents: 8 associated with NVIDIA, 8 with AMD, and 4 with Microsoft. These documents included primary 8-K filings and supporting exhibits.

The collected SEC documents were downloaded and parsed using BeautifulSoup. HTML markup, unnecessary whitespace, and non-content elements were removed to produce readable text as well.

Because SEC filings vary significantly in length, documents were divided into overlapping segments of approximately 5,000 characters, with a 500-character overlap. This approach was mainly to help accommodate the LLM's context-window limitations while reducing the likelihood of missing information across segment boundaries.

The extraction process used Meta's Llama 3.2 3B model, deployed through Ollama in a Google Colab environment with an NVIDIA Tesla T4 GPU.

The model was queried programmatically through Ollama's local HTTP API. Prompt engineering was used to guide the model in performing entity recognition and identifying explicit product launches, introductions, and announcements.

The instructions distinguished new product announcements from references to existing products, financial performance, and unrelated corporate events. A temperature setting of zero was used to encourage more consistent outputs.

A predefined JSON schema structured the model responses, including product names, short descriptions, and supporting evidence extracted from the documents.

The extracted product information was converted into a Pandas DataFrame containing the five required fields:

company_name, stock_name, filing_time, new_product, product_description

Duplicate product records within the same filing were removed based on company ticker, filing timestamp, and product name. Product descriptions were restricted to fewer than 180 characters, and the resulting structured dataset was exported as a UTF-8 CSV file.

The initial implementation successfully retrieved SEC filing metadata, parsed filing documents and supporting exhibits, and generated structured product information using the LLM.

During an initial NVIDIA exhibit analysis, the model identified several product-related entities, including NVIDIA Vera, NVIDIA DSX, and NVIDIA BioNeMo Agent Toolkit. This demonstrated the feasibility of extracting product information from lengthy, unstructured corporate disclosures. However, the initial extraction also identified products associated with production milestones and benchmark announcements. This highlighted an important limitation that showed that recognizing a product entity does not necessarily establish that a new product has been announced.

Several limitations were identified. First, not all Form 8-K filings contain product announcements, so the extracted dataset may be considerably smaller than the number of documents processed. Second, smaller LLMs can incorrectly classify existing products as new releases. Third, processing long documents in separate segments may cause some contextual information to be lost.

Additionally, the project did not include a manually labeled benchmark dataset, so formal precision and recall measurements were not calculated in this project.

Overall, the project demonstrates how LLMs can support financial document analysis by transforming unstructured regulatory disclosures into structured data. The resulting pipeline combines publicly available SEC data, document preprocessing, NER-oriented prompting, and automated CSV generation. Future improvements could include human-reviewed evaluation datasets, larger language models, and more extensive coverage of S&P 500 companies.